PSY 5790: Applied Univariate and Bivariate/Bivariable Statistics

Correlation and Simple Linear Regression Demo: Continuous Predictor

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Required Libraries

```
library(tidyverse)
library(dplyr)
library(kableExtra)
library(psych)
library(jmv)
library(lm.beta)
library(skimr)
library(broom)
library(moderndive)
library(gapminder)
```

First load in the data

```
load(url("http://www.openintro.org/stat/data/evals.RData"))
evals <- evals %>%
  select(score, bty_avg, age)
```

Exploratory Data Analysis

1. Observe raw data
2. Compute descriptive statistics
3. Visualize data

Observe raw data

Remember, you should always technically do more than `head()` or `glimpse()` — take the time to observe the raw data to see if anything jumps out at you!

```
head(evals)
```

```
##   score bty_avg age
## 1   4.7       5  36
## 2   4.1       5  36
## 3   3.9       5  36
## 4   4.8       5  36
## 5   4.6       3  59
## 6   4.3       3  59
```

```
#glimpse(evals)
#View(evals)
```

Compute descriptive statistics

```
## Can use any of the function calls below to conduct descriptives
## My personal preference is the `describe()` function
describe(evals)
```

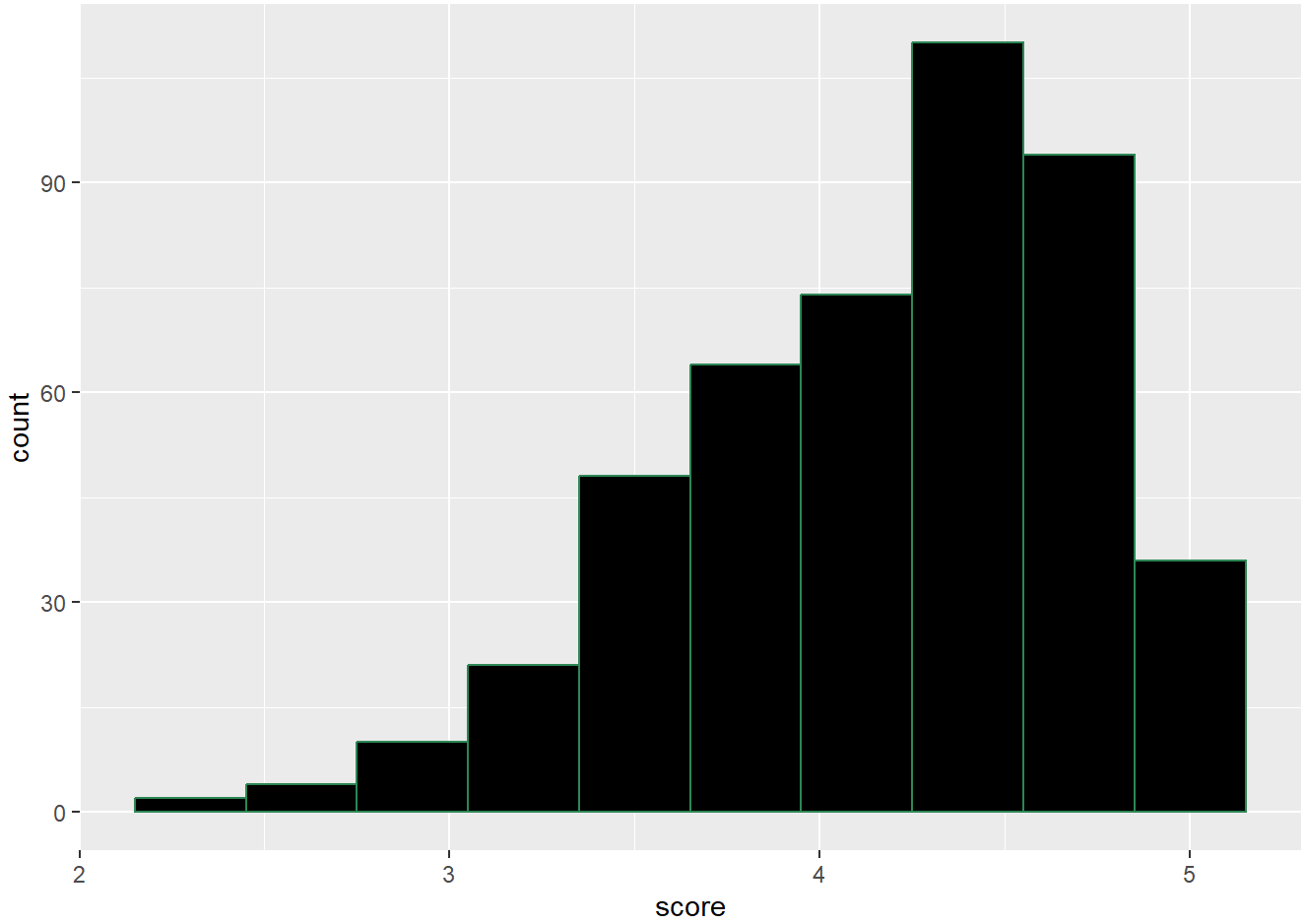
```
##         vars   n  mean   sd median trimmed   mad   min   max range  skew
## score      1 463  4.17 0.54   4.30    4.22  0.59  2.30  5.00   2.7 -0.70
## bty_avg    2 463  4.42 1.53   4.33    4.33  1.73  1.67  8.17   6.5  0.50
## age        3 463 48.37 9.80  48.00   48.35 11.86 29.00 73.00  44.0  0.05
##         kurtosis   se
## score       0.04 0.03
## bty_avg    -0.43 0.07
## age        -0.80 0.46
```

```
## These alternatives give you many descriptive stats but nothing about normality
# skim(evals)
# summary(evals)
# skim_without_charts(evals)

## Don't forget correlations!
corrMatrix(evals)
```

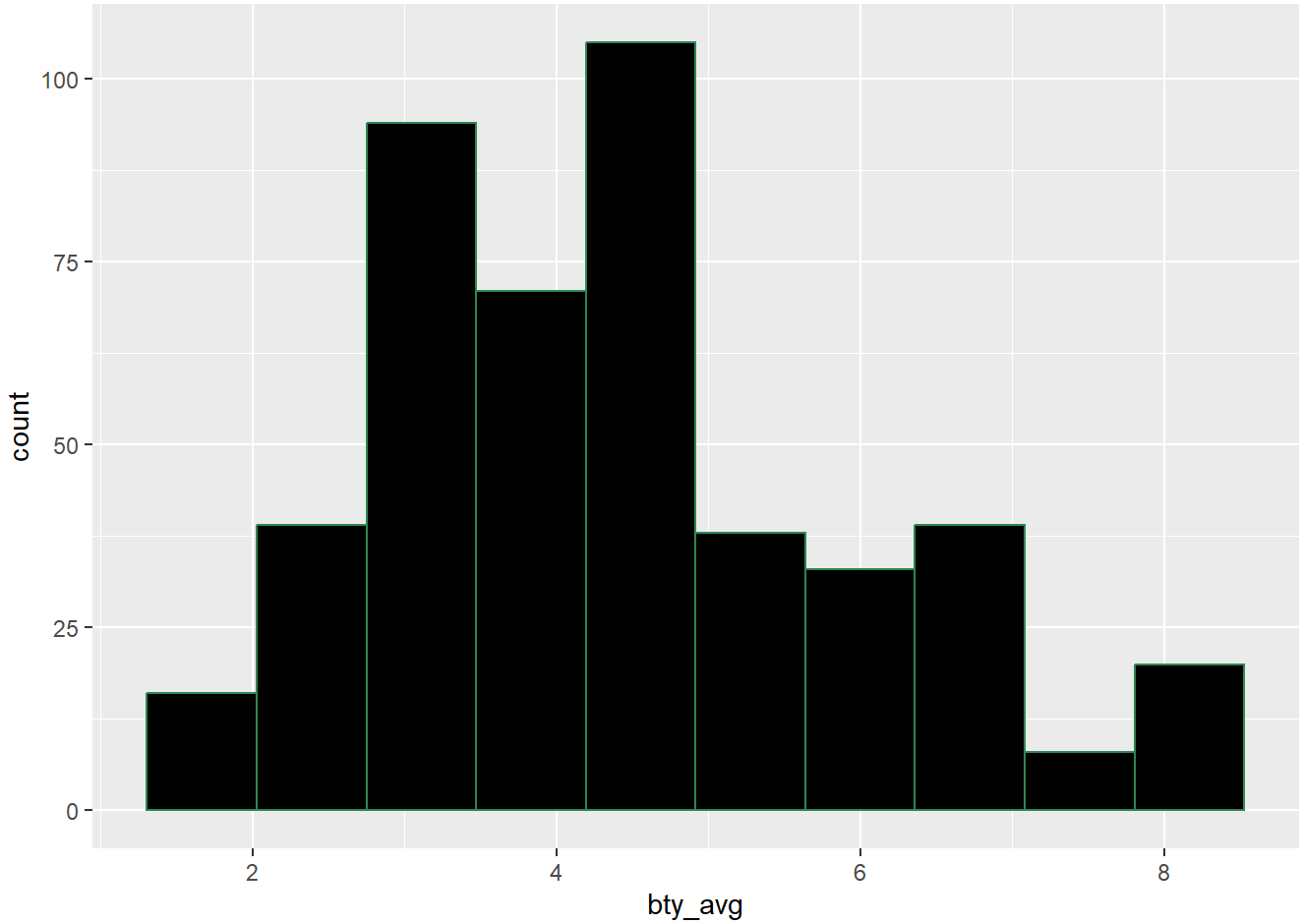
```
##
##  CORRELATION MATRIX
##
##  Correlation Matrix
##  ────────────────────────────────────────────────────────────────
##                              score         bty_avg       age
##  ────────────────────────────────────────────────────────────────
##    score      Pearson's r             —
##               df                      —
##               p-value                 —
##
##    bty_avg    Pearson's r     0.1871424             —
##               df                    461             —
##               p-value         0.0000508             —
##
##    age        Pearson's r    -0.1070320    -0.3046034            —
##               df                    461           461            —
##               p-value         0.0212536    < .0000001            —
##  ────────────────────────────────────────────────────────────────
```

Visualize the data

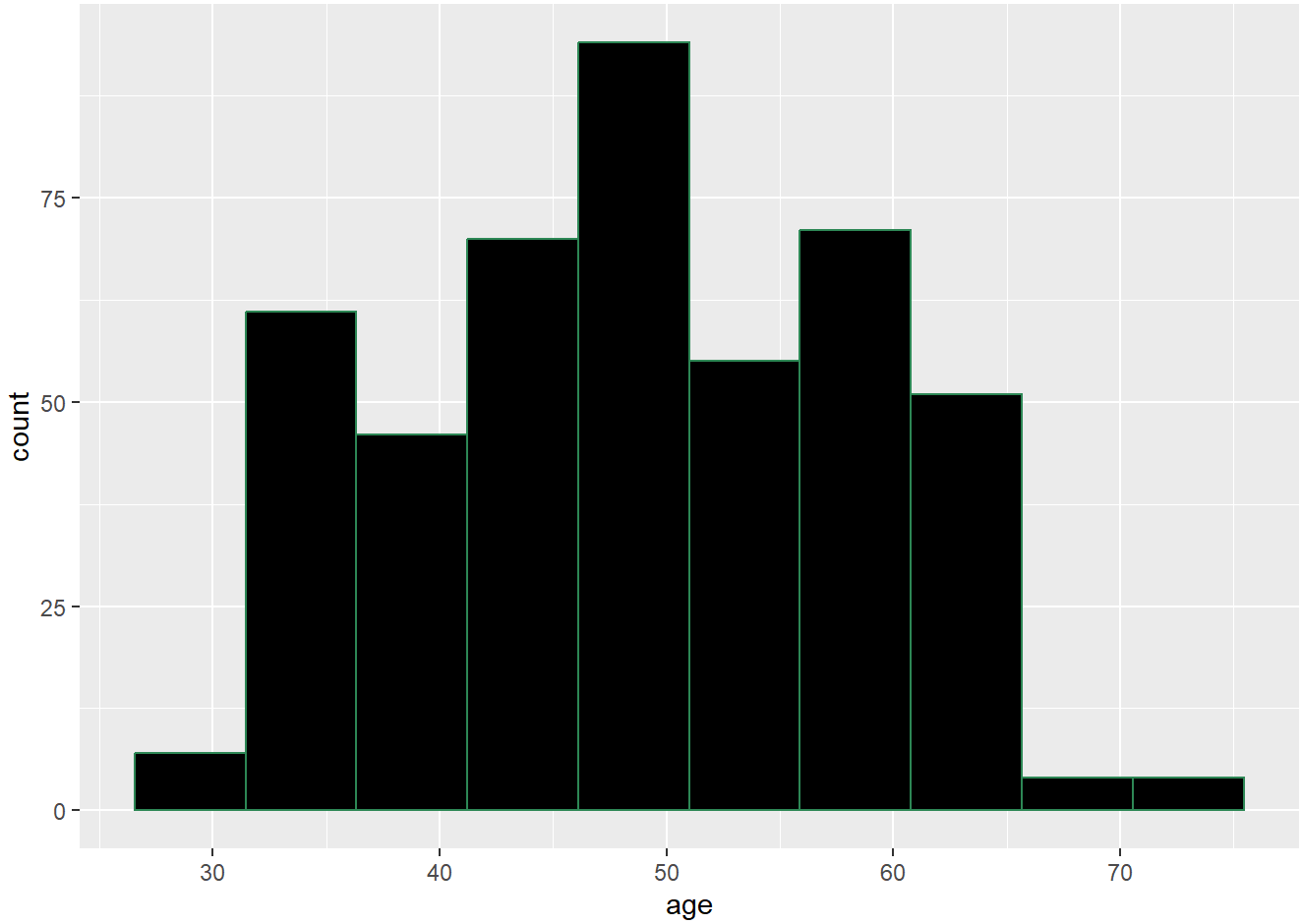
```
## Histograms

ggplot(data = evals,
       mapping = aes(x = score)) +
  geom_histogram(bins = 10,
                 color = "seagreen",
                 fill="black")
```



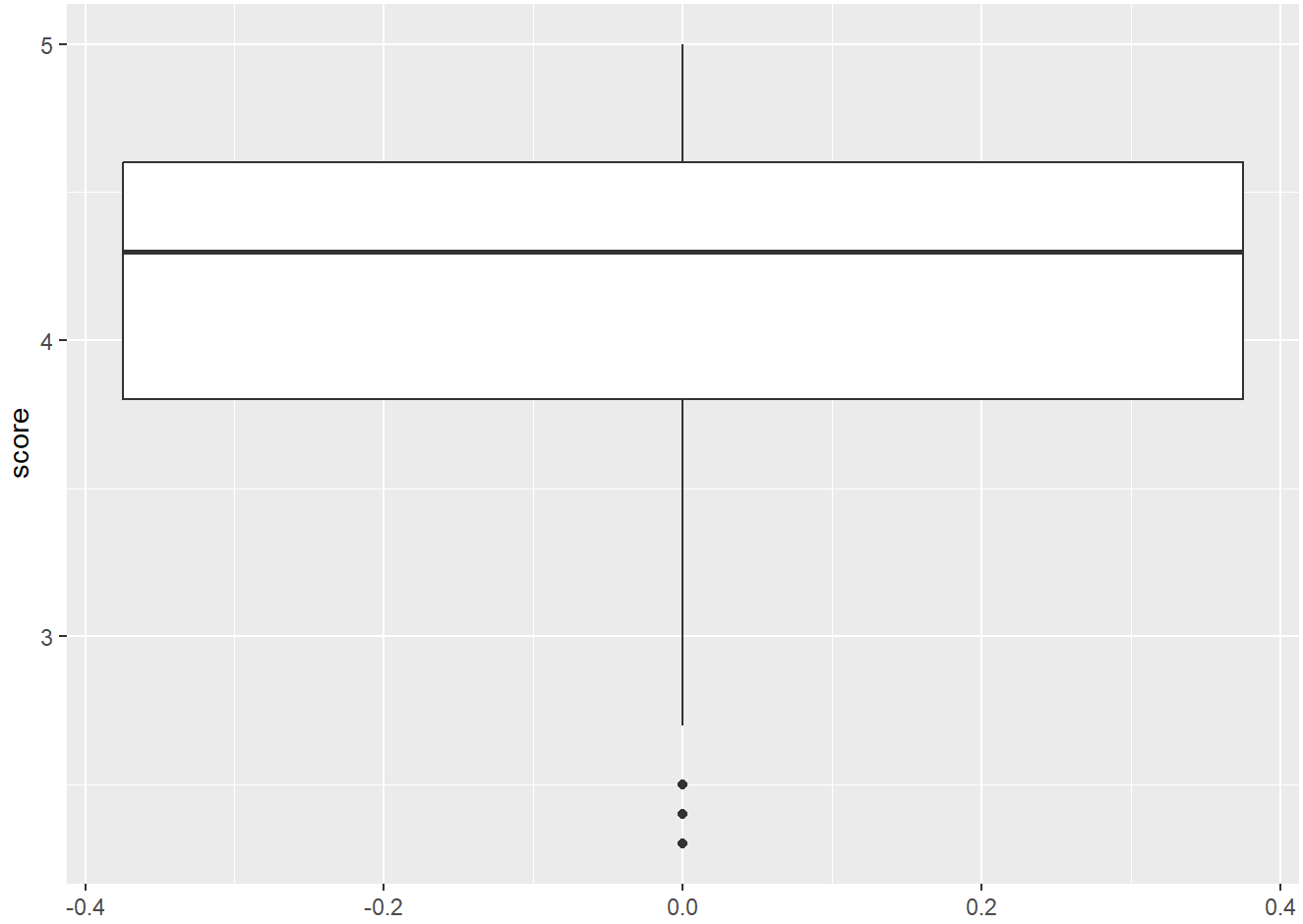

```
ggplot(data = evals,
       mapping = aes(x = bty_avg)) +
  geom_histogram(bins = 10,
                 color = "seagreen",
                 fill="black")
```



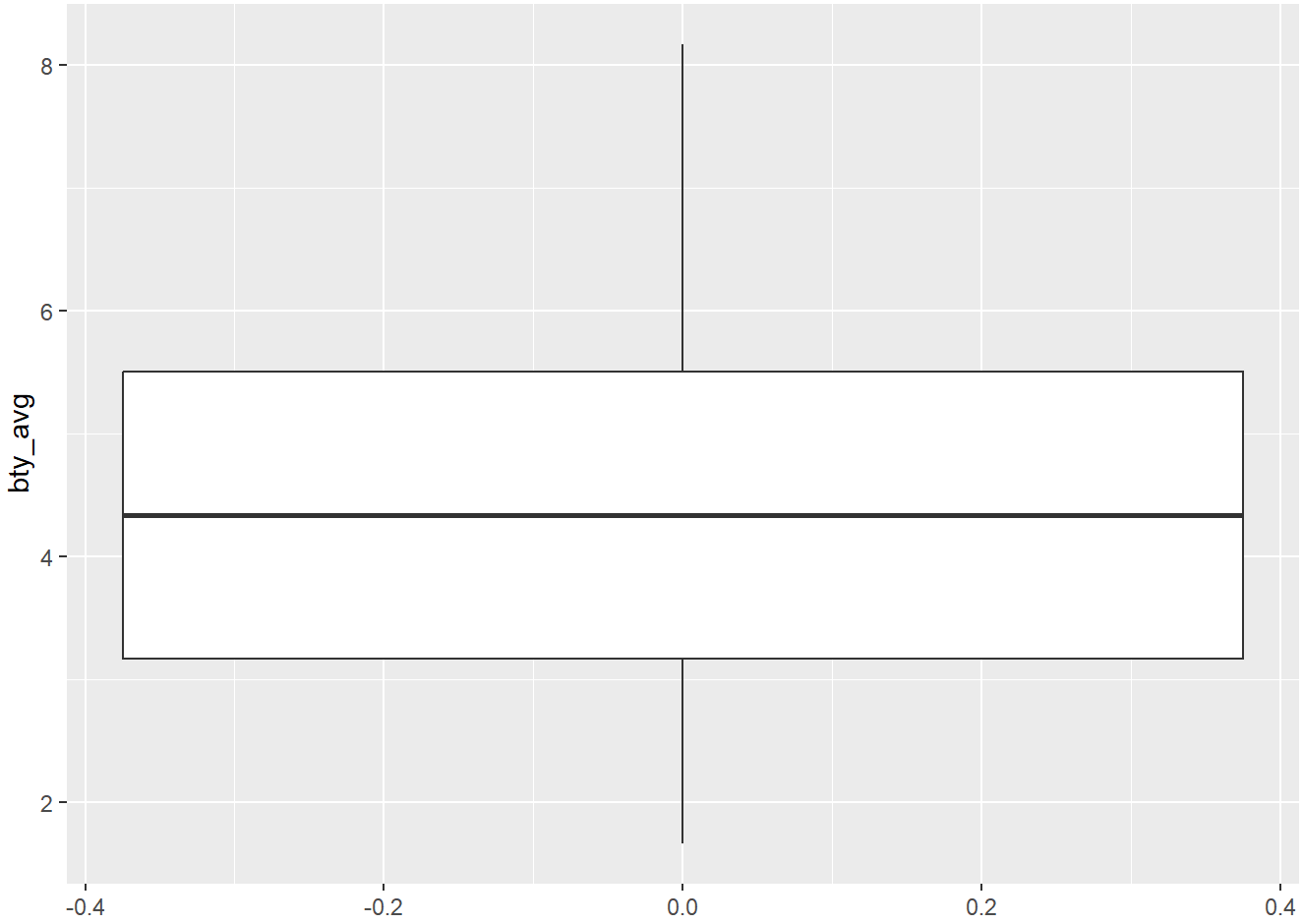

```
ggplot(data = evals,
       mapping = aes(x = age)) +
  geom_histogram(bins = 10,
                 color = "seagreen",
                 fill="black")
```



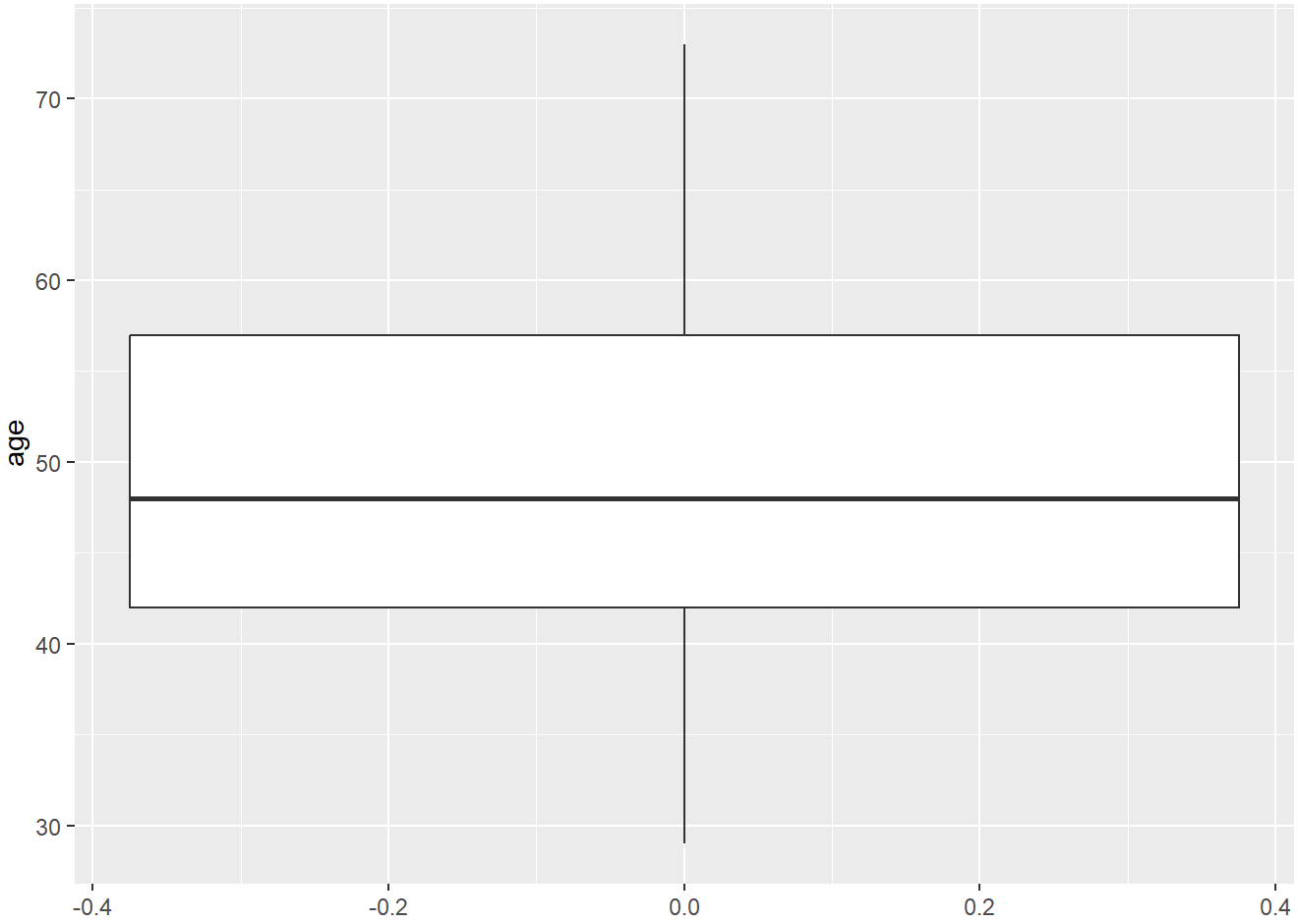

```
## Boxplots
ggplot(data = evals,
       mapping = aes(y = score)) +
  geom_boxplot()
```



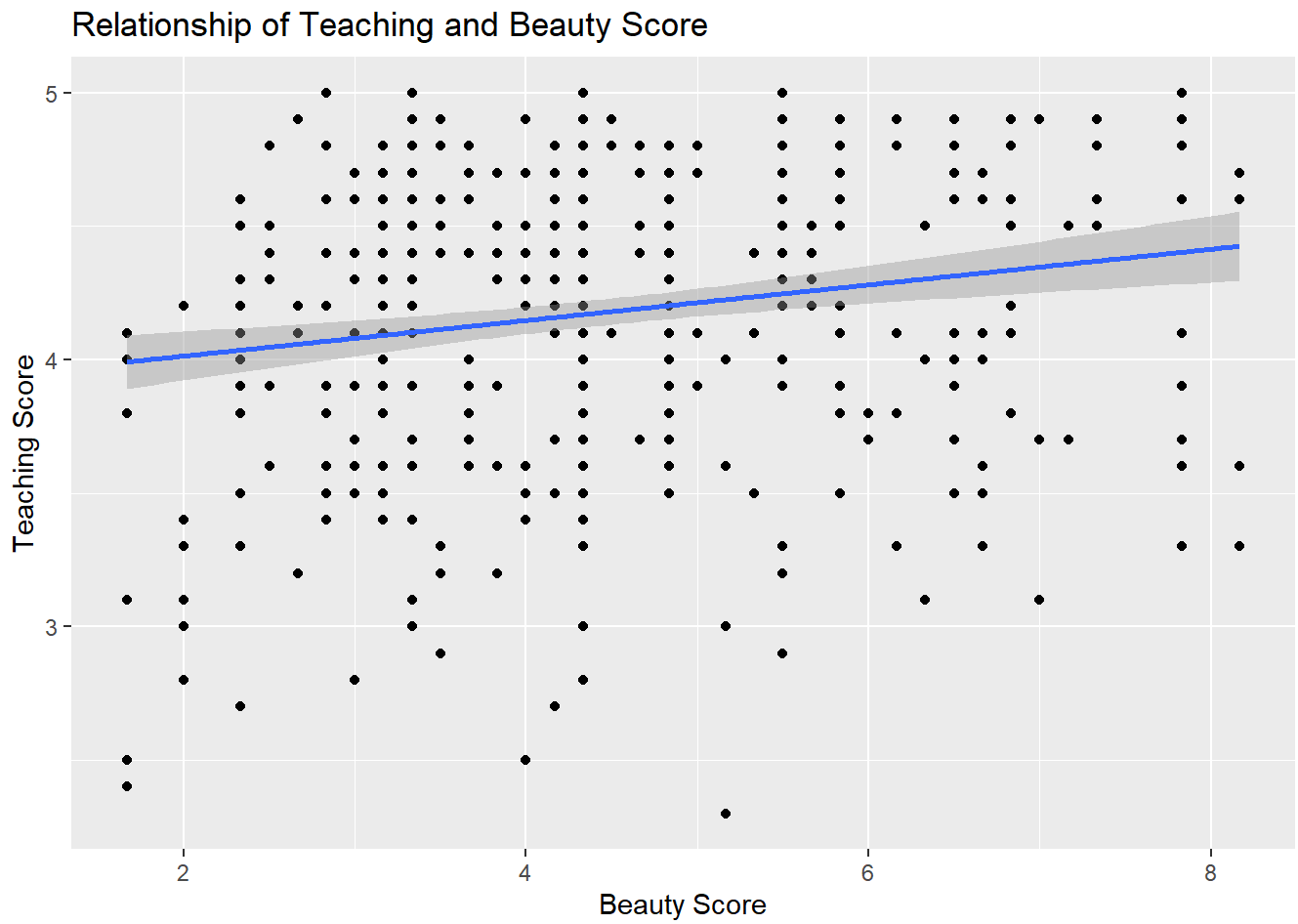

```
ggplot(data = evals,
       mapping = aes(y = bty_avg)) +
  geom_boxplot()
```



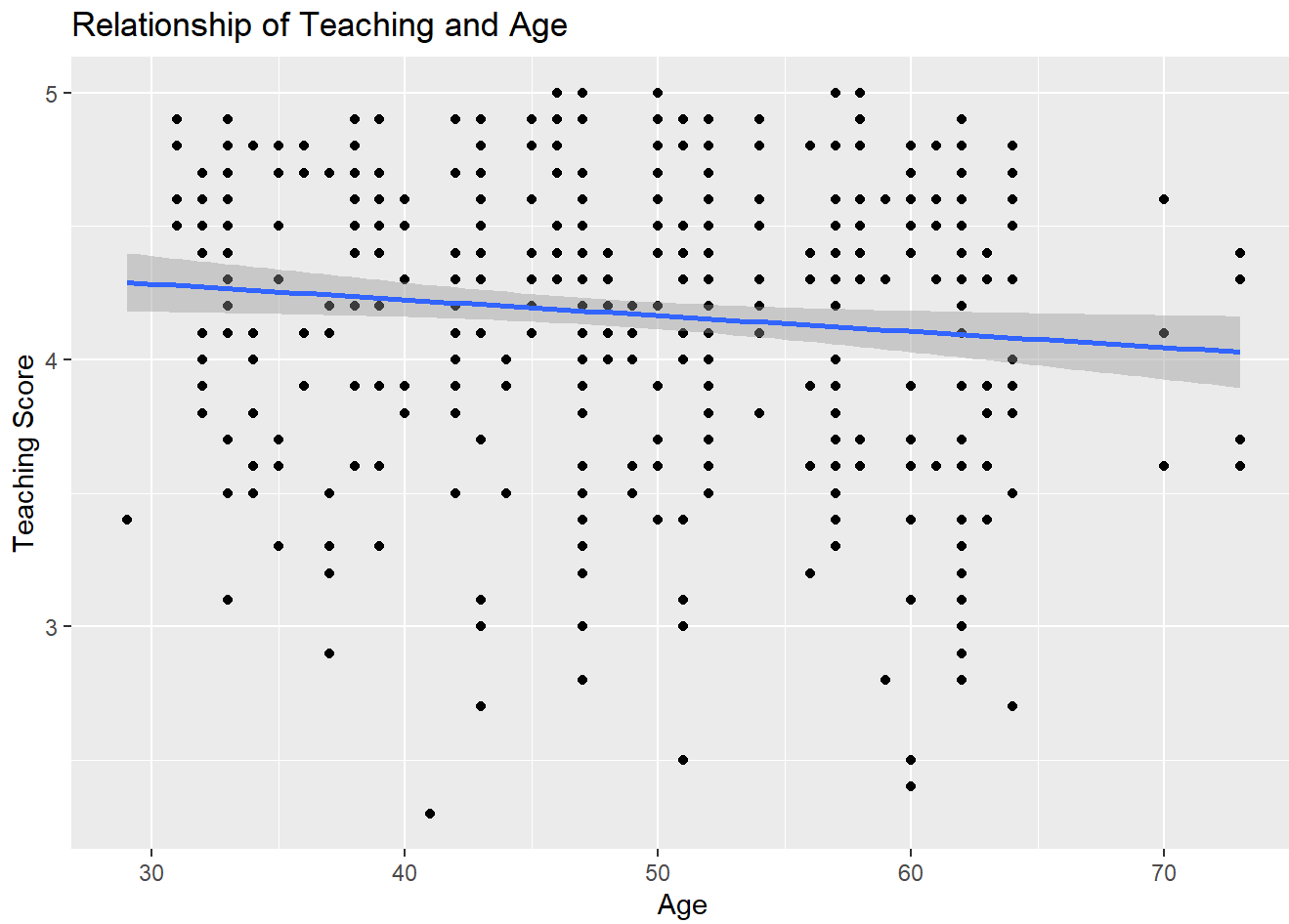

```
ggplot(data = evals,
       mapping = aes(y = age)) +
  geom_boxplot()
```




```
## Scatterplots
ggplot(evals,
       aes(x = bty_avg,
           y = score)) +
  geom_point() +
  labs(x = "Beauty Score",
       y = "Teaching Score",
       title = "Relationship of Teaching and Beauty Score") +
  geom_smooth(method = "lm")
```




```
ggplot(evals,
       aes(x = age,
           y = score)) +
  geom_point() +
  labs(x = "Age",
       y = "Teaching Score",
       title = "Relationship of Teaching and Age") +
  geom_smooth(method = "lm")
```



Now lets run our simple linear regression (`lm()` workflow + `broom` package)

```
score_model <- lm(score ~ bty_avg, data = evals)

# Get regression table using `tidy()`:
tidy(score_model, conf.int=TRUE)
```

```
## # A tibble: 2 × 7
##   term        estimate std.error statistic   p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
## 1 (Intercept)   3.88      0.0761     51.0  1.56e-191   3.73      4.03
## 2 bty_avg       0.0666    0.0163      4.09 5.08e-  5   0.0346    0.0987
```

```
# Get model fit table using `glance()`:
glance(score_model)
```

```
## # A tibble: 1 × 12
##   r.squared adj.r.squared sigma statistic   p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>     <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1    0.0350        0.0329 0.535      16.7 0.0000508     1  -366.  738.  751.
## # ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
# Get Standardized Regression coefficients
lm.beta(score_model)
```

```
##
## Call:
## lm(formula = score ~ bty_avg, data = evals)
##
## Standardized Coefficients::
## (Intercept)     bty_avg
##          NA   0.1871424
```

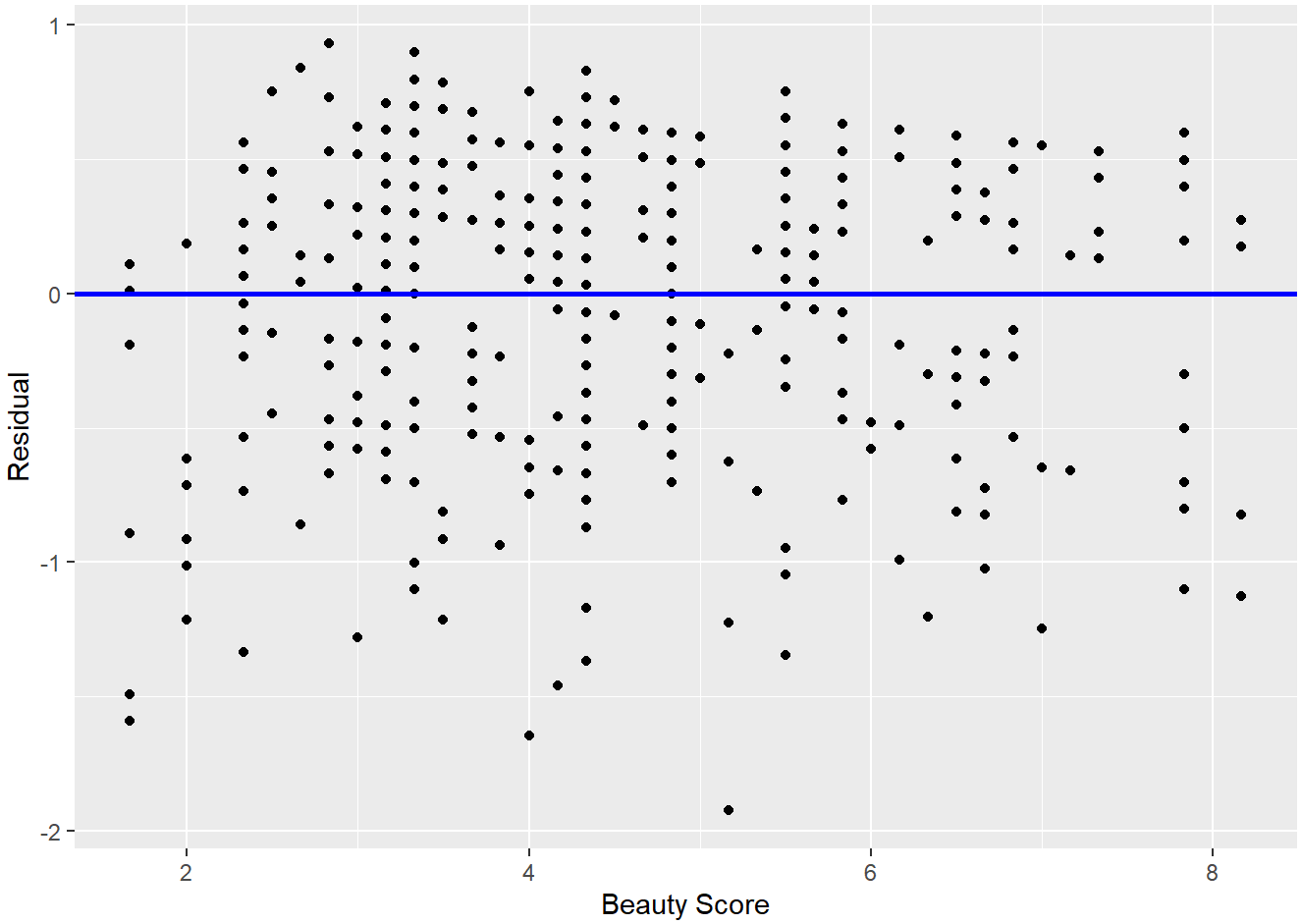
Alternatively we can use `linReg()` from the `jmv` package

```
linReg(evals,
       dep=score,
       covs=bty_avg,
       blocks=list(list('bty_avg')),
       anova=TRUE,
       stdEst=TRUE,
       ci=TRUE,
       ciStdEst=TRUE)
```

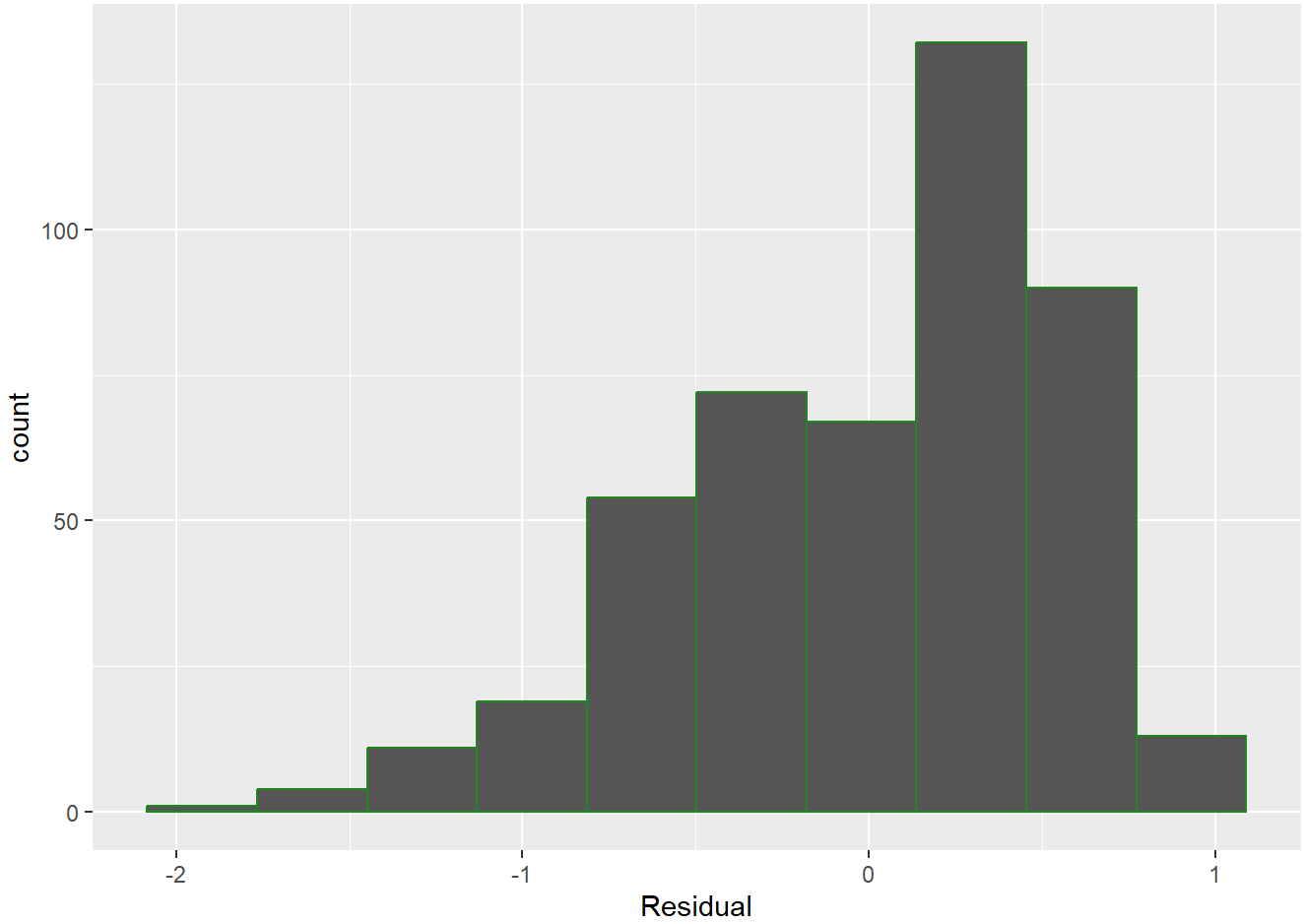
```
##
##  LINEAR REGRESSION
##
##  Model Fit Measures
##  ─────────────────────────────────────
##    Model    R            R²
##  ─────────────────────────────────────
##        1    0.1871424    0.03502226
##  ─────────────────────────────────────
##    Note. Models estimated using
##    sample size of N=463
##
##
##  MODEL SPECIFIC RESULTS
##
##  MODEL 1
##
##  Omnibus ANOVA Test
##  ──────────────────────────────────────────────────────────────────────────────
##                 Sum of Squares    df     Mean Square    F           p
##  ──────────────────────────────────────────────────────────────────────────────
##    bty_avg            4.785944      1      4.7859440    16.73123    0.0000508
##    Residuals        131.868397    461      0.2860486
##  ──────────────────────────────────────────────────────────────────────────────
##    Note. Type 3 sum of squares
##
##
##  Model Coefficients - score
##  ──────────────────────────────────────────────────────────────────────────────
```

Lastly we should conduct a residual analysis

```
regression_points <- get_regression_points(score_model)

# Scatterplot of Explanatory Variable along x-axis and Residuals along y-axis
ggplot(regression_points,
       aes(x = bty_avg, y = residual)) +
  geom_point() +
  labs(x = "Beauty Score", y = "Residual") +
  geom_hline(yintercept = 0, col = "blue", size = 1)
```




```
# Additionally we want to look at the distribution of residuals
ggplot(regression_points,
       aes(x = residual)) +
  geom_histogram(bins = 10, color = "forestgreen") +
  labs(x = "Residual")
```




```
describe(regression_points$residual)
```

```
##    vars   n mean   sd median trimmed  mad   min  max range  skew kurtosis   se
## X1    1 463    0 0.53   0.14    0.04 0.54 -1.93 0.93  2.86 -0.71    -0.01 0.02
```